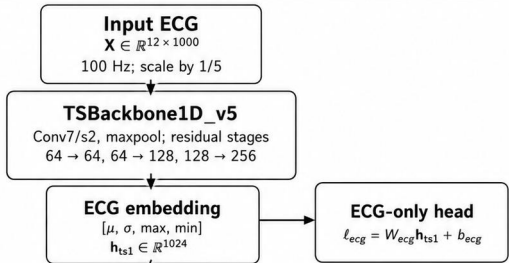
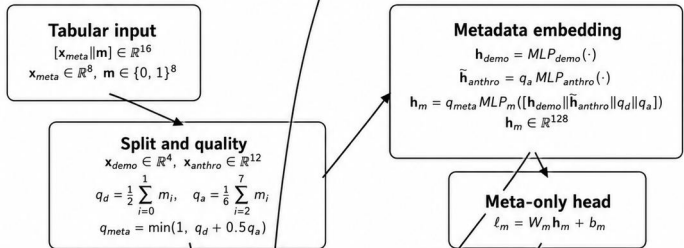


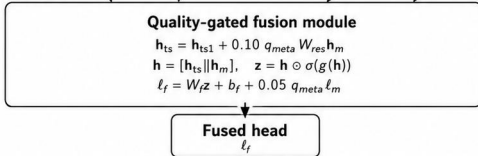
ECG branch



Metadata input



Cross-modal fusion



Training objective

$$0.52 \text{ BCE}_w + (\ell_f, y) + 0.30 \text{ BCE}_w + (\ell_{ecg}, y) + 0.10 q_{meta} \text{ BCE}(\ell_m, y) + 0.08 \mathcal{L}_{AUC}(\sigma(\ell_f), y)$$

Fixed-a-priori inference

$$\mathbf{p} = \sigma(\ell_f) \quad (w^* = 1.0 \text{ fixed a priori}); \quad \text{class thresholds tuned on validation}$$